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Fuzzy Password Authentication Key Exchange protocol in universal composable framework for blockchain privacy protection

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ABSTRACT

In this paper, we construct a lattice-based fuzzy Password Authentication Key Exchange protocol in universal composable model. Through the known Password Authentication Key Exchange scheme, the Randomized Fuzzy Equality protocol and the Oblivious Transfer protocol are introduced to improve Password Authentication Key Exchange into fuzzy Password Authentication Key Exchange. First, the parties go through two rounds of Oblivious Transfer protocol, and then the key exchange is achieved based on the information exchanged. fuzzy Password Authentication Key Exchange satisfies that even if there is noise in the passwords between users, key exchange is still possible. Therefore, fuzzy Password Authentication Key Exchange is suitable for more application scenarios compared to Password Authentication Key Exchange, and the construction is universal composable security.

1. Introduction

Authentication is essential for establishing a shared key among two communicating entities. In the process, the user employs a password as an authentication credential, exchanging messages within a protocol to generate the shared key. However, more complex scenarios may emerge during this interaction. For instance, passwords may be derived from heterogeneous, low-entropy distributions. Password inputs from both parties may include noise, resulting in potential mismatches. To address these issues, the fuzzy Password Authentication Key Exchange (fPAKE) protocol [1] was developed based on the Password Authentication Key Exchange (PAKE) protocol [2]. Traditional PAKE protocols require both parties to input identical passwords. However, when parties are faced with constraints of password policy, the noise may arise between their passwords. The fPAKE protocol alleviates the stringent password requirements inherent in traditional PAKE system.

The combination of cryptography protocols and blockchain plays an crucial role in network security. Cryptography protocols provide a method for securing the transmission of information. Blockchain is a decentralized, distributed database technology that enables transparency, immutability, and decentralization of information [3]. Blockchain allows parties to verify the authenticity of transactions by recording details in a chronologically ordered chain structure [4]. Cryptography protocols can be employed to protect various forms of communication, such as email, instant messaging, and online banking. However, traditional cryptography protocols face several limitations. For instance, centralized authentication mechanisms are susceptible to attacks and forgery. The integration of cryptography protocols with blockchain can effectively mitigate identity forgery and impersonation, as authentication information is recorded on the blockchain [5–8]. PAKE as commonly security protocol that supports authentication and key exchange over unsecured communication channels. The combination of PAKE and blockchain improves the integrity and trustworthiness for key exchange protocols. Unlike PAKE, the fPAKE protocol does not impose strict requirements on user passwords, indicating that the fPAKE protocol holds significant potential for various application scenarios.

In 1988, Bennett et al. [15] proposed the concept of fuzzy extractor. In 1992, Bellovin and Merritt [16] proposed a basic password authentication protocol based on Diffie–Hellman key exchange. Then, the password authentication mechanism was introduced. In 1998, Thomas Wu [17] proposed a PAKE protocol called Secure Remote Password (SRP). The SRP protocol uses the technique of zero-knowledge proof, which makes it impossible for the server to obtain the user's password. The SRP protocol was one of the milestones of the PAKE protocol with significant improvements in security and usability. In 2000, Victor Boyko et al. [18] provided a more rigorous security proof for PAKE by introducing a formal approach to cryptography. In 2004, Regev [19] first introduced the Learning With Errors (LWE) problem in lattice-based cryptography and demonstrated that it can resist quantum attacks. In 2006, Goldreichd et al. [20] proposed the first PAKE protocol that is provable security under the Standard model, but the communication and computational cost of the protocol is relatively

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Table 1
Summary of past results from PAKE.

Work	Fuzzy	Publication year	Summary
[1]	✓	2018	Examined the key agreement from noisy low-entropy pass-strings and the method of handle low-entropy pass-strings.
[8]	*	2019	Examined the application of hash functions and cryptography at different levels of the blockchain, including the data layer, the network layer and the consensus layer.
[9]	✓	2020	Examined interactivity between parties, and the size of message and password files are made independent of the fault tolerance.
[10]	×	2021	A lattice-based anonymous PAKE supporting key reuse is proposed, but is not UC security.
[11]	×	2022	A lattice-based three-party oriented PAKE protocol with low communication and computation cost is proposed.
[12]	✓	2023	Examined party's personal password protection mechanism and provide a more security fPAKE protocol.
[13]	×	2023	Examined IBE-based PAKE protocol, where IBE is provable safe.
[14]	×	2023	Examined asymmetric PAKE protocol on IoT and a provable security aPAKE protocol is presented.

large. In 2008, Dodis et al. [21] employed the Hamming distance to achieve key exchange between parties, assuming the noise in their passwords remained within a tolerable range. In 2009, Katz et al. [22] introduced the first lattice-based PAKE protocol, which is primarily based on the approximate Smooth Projective Hash Function (SPHF). In 2010, Groce et al. [23] proposed the Groce–Katz framework that consists of the IND-CCA security public-key algorithm and SPHF.

1.1. Related work

In recent years, lattice-based PAKE protocols [10,11,24] have received increasing attention. You Lyu et al. [25] proposed a PAKE protocol with bidirectional authentication at EUROCRYPT, whose security is based on the LWE problem, and demonstrated that their protocol achieved UC security. Furthermore, there has been a surge of research on PAKE protocols and their applications. As a variant of PAKE, fPAKE has received relatively less research attention compared to PAKE. Recently, there has also been growing research on both PAKE protocols [13,14] and fPAKE protocols [1,9,12].

As shown in Table 1, in 2018, Dupont, P.A. et al. [1] presented a study on key agreement protocols based on noisy low-entropy passwords. The authors proposed a UC-secure fPAKE protocol whose security is based on bilinear pairings. In 2019, Zhai, S. et al. [8] discussed the application of various cryptographic primitives to blockchain, providing a detailed description of hash functions. In 2020, Erwig et al. [9] introduced a fuzzy asymmetric PAKE protocol, where message and password files are independent of fault tolerance. The protocol enhances the efficiency of interactions between parties and resists malicious attacks. In 2021, Dabra, V. et al. [10] introduced a lattice-based anonymous PAKE supporting key reuse with perfect forward secrecy, though without UC-secure, addressing an earlier security vulnerability. Most previous PAKE protocols are designed for two-party interactions. In 2022, Xie, H. et al. [11] introduced a lattice-based three-party PAKE protocol with reduced communication and computation costs. In 2023, Bootle, J. et al. [12] proposed a password protection mechanism that offers a stronger security proof. Priyadharshini, A. et al. [13] proposed an Identity-Based Encryption (IBE)-based PAKE protocol for identity access control, with the specific implementation steps provided in their article. Qi, M. et al. [14] introduced an asymmetric PAKE protocol for IoT applications, with security based on bilinear pairings.

Most current fPAKE protocols calculate noise in passwords from different parties using the Hamming distance. If the calculated Hamming distance is below a certain threshold, a correct key exchange can be performed between the communicating parties. The existence of the fPAKE protocol allows for key exchanges despite noise in parties' passwords, provided that the passwords differ within a specific distance. Therefore, the fPAKE protocol is particularly suited for complex network environments. In blockchain environments, where conditions are often complex, the fPAKE protocol is well-suited for facilitating secure key exchanges. Since transaction information is typically public in blockchain systems, this openness may lead to the leakage of user privacy. The UC-secure protocol can be employed to design and implement privacy protection mechanisms. Moreover, the UC-secure fPAKE protocol ensures the privacy of transactions and prevents the leakage of sensitive information, thereby enhancing privacy protection in blockchain.

1.2. Our contributions

PAKE protocols primarily serve to authenticate user identity and enable secure key exchange. The security of PAKE heavily relies on the strength of the password; the use of weak passwords significantly reduces the protocol's security. In traditional PAKE protocols, both parties are required to use the same password for communication, making the protocol susceptible to password guessing attacks.

(1) This work primarily modifies the PAKE protocol from [25] by combining the Randomized Fuzzy Equality (RFE) protocol and the OT protocol to achieve a UC-secure fPAKE protocol. In the proposed fPAKE protocol, for fuzzy passwords between the two parties, the information is first exchanged using the OT protocol. The RFE protocol is then combined to verify the Hamming distance between the fuzzy passwords, ensuring security verification. Compared to [1,25], this work proposes a more secure and versatile fPAKE protocol. Since the proposed protocol is UC-secure, it is secure for use when combined with other protocols.
(2) This paper constructs the fPAKE protocol based on the PAKE protocol. The proposed fPAKE protocol is based on the LWE hard problem, whereas most existing fPAKE protocols rely on bilinear pairings.
(3) When applied to the Internet of Vehicles (IoV), the proposed fPAKE protocol enables more flexible access control to vehicles. In terms of both application and security, fuzzy password authentication of the proposed fPAKE protocol can be securely facilitates flexible access control.

1.3. Organization

In Section 2, the basics of lattice theory, along with several lattice-related definitions and algorithms employed in this work, are introduced, including the LWE hard problem. Section 3 presents the cryptographic primitives utilized in this paper, such as the RFE protocol and the OT protocol. In Section 4, the focus is on the UC-secure framework and ideal functionalities within the security proofs. Section 5 presents the UC-secure fPAKE protocol. Section 6 provides a performance analysis of the proposed protocol. Section 7 outlines an application scenarios, and Section 8 provides a summary of the proposed protocol.

Table 2
Notions in this paper.

Notation	Mean	Notation	Mean
$\mathbb{Z}$	The ring of the integers and the real numbers	$a \leftarrow F(x)$	The input x outputs a through the algorithm F
$\mathbb{Z}_q^{n\times m}$	The set of $n \times m$ matrices with entries in $\mathbb{Z}_q$	$\mathcal{A}$	Adversary in Security Proof
$\boldsymbol{A}$	Bold capitals denote matrix	$\mathcal{Z}$ or Evn	The Environment in Security Proof
$\boldsymbol{a}$	Lowercase bold denotes vector	π	The protocol of this paper
$\|\boldsymbol{A}\|$	The Euclidean norm and the transposed matrix of $\boldsymbol{A}$	Sim	Simulator in Security Proof
$\boldsymbol{A}^{\mathrm{T}}$	The transposed matrix of $\boldsymbol{A}$	$\mathcal{P}_i$	Denotes one of the two parties
$x \leftarrow \chi$	From the set χ uniformly with randomly selected x	pw_i	Denotes the password of one of the parties
$\|\chi\|$	Number of elements in set χ	$negl(n)$	Negligible function of n

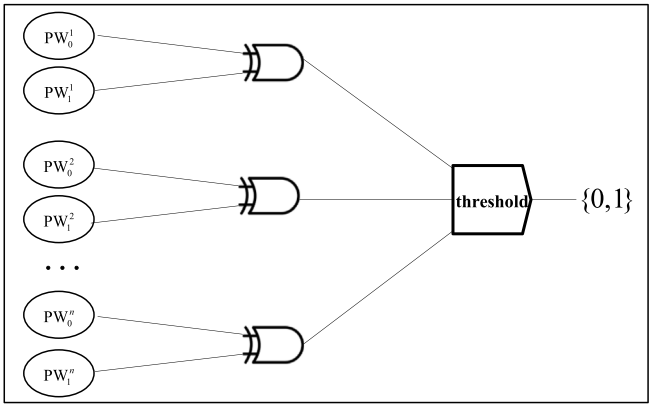


Fig. 1. A circuit f.

2. Fundamentals

In this subsection, we briefly introduce the mathematical concepts covered in the this paper. And the notation involved in this paper as shown in Table 2.

2.1. Lattice

Definition 1. Let $\boldsymbol{b}_1, \boldsymbol{b}_2 \cdots \boldsymbol{b}_m \in \mathbb{R}^m$ be a set of linearly independent vectors, and a m-dimensional full rank lattice [26]:

$$\Lambda = L\left(\boldsymbol{b}_1, \boldsymbol{b}_2 \cdots \boldsymbol{b}_m\right) = \left\{ \boldsymbol{y} \in \mathbb{R}^m, s.t. \exists \boldsymbol{s} \in \mathbb{Z}^m, \boldsymbol{y} = \sum_{i=1}^{m} \boldsymbol{s}_i \boldsymbol{b}_i \right\} \tag{1}$$

Definition 2. Let q be a prime, a matrix $\boldsymbol{A} \in \mathbb{Z}_q^{n\times m}$ and a vector $\boldsymbol{u} \in \mathbb{Z}_q^n$, define three m-dimensional full rank lattices [27]:

$$\Lambda_q(\boldsymbol{A}) = \left\{ \boldsymbol{e} \in \mathbb{Z}^m, s.t. \exists \boldsymbol{s} \in \mathbb{Z}_q^n, \boldsymbol{A}^{\mathrm{T}}\boldsymbol{s} = \boldsymbol{e} \bmod q \right\} \tag{2}$$

$$\Lambda_q^{\perp}(\boldsymbol{A}) = \{ \boldsymbol{e} \in \mathbb{Z}^m, s.t. \boldsymbol{A}\boldsymbol{e} = 0 \bmod q \} \tag{3}$$

$$\Lambda_q^{u}(\boldsymbol{A}) = \{ \boldsymbol{e} \in \mathbb{Z}^m, s.t. \boldsymbol{A}\boldsymbol{e} = \boldsymbol{u} \bmod q \} \tag{4}$$

If $t \in \Lambda_q^{u}(\boldsymbol{A})$, then $\Lambda_q^{u}(\boldsymbol{A}) = \Lambda_q^{\perp}(\boldsymbol{A}) + t$.

Lemma 1 ([28]). *Let a prime $q \geq 3$ and a positive integer $m = \lfloor 6n \log q \rfloor$. There is a PPT (Probabilistic Polynomial Time) algorithm $\boldsymbol{TrapGen}(q, n)$ that output $\boldsymbol{A} \in \mathbb{Z}_q^{n\times m}$ and $\boldsymbol{T}_{\boldsymbol{A}} \in \mathbb{Z}^{m\times m}$, where the matrix $\boldsymbol{A}$ is statistically close to uniform distribution and $\boldsymbol{T}_{\boldsymbol{A}}$ is a basis for $\Lambda_q^{\perp}(\boldsymbol{A})$.*

Definition 3 ([29]). For any vector $\boldsymbol{c} \in \mathbb{R}^n$ and a positive parameter $s > 0$, the Gaussian distribution function on $\mathbb{R}^n$:

$$\forall \boldsymbol{x} \in \mathbb{R}^n, \rho(\boldsymbol{x}) = e^{-\pi\|(\boldsymbol{x}-\boldsymbol{c})/s\|^2} \tag{5}$$

Discrete Gaussian distribution function is defined over m-dimensional lattice Λ:

$$D_{\Lambda,s,\boldsymbol{c}}(\boldsymbol{x}) = \frac{\rho_{s,\boldsymbol{c}}(x)}{\rho_{s,\boldsymbol{c}}(\Lambda)} \tag{6}$$

2.2. LWE hard problem

The LWE problem is to give LWE's pair and uniform random distribution, and distinguishing between them is difficult.

- **Decision LWE:** For some samples $(\boldsymbol{A}, \boldsymbol{c}) = (\boldsymbol{A}, \boldsymbol{A}^{\mathrm{T}}\boldsymbol{s}+\boldsymbol{x}) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m$, where the distribution of $\boldsymbol{A} \leftarrow \mathbb{Z}_q^{n\times m}$ and $\boldsymbol{s} \leftarrow \mathbb{Z}_q^n$ are chosen from a uniform distribution, $\boldsymbol{x}$ is randomly sample. Finding $\boldsymbol{x}$ is difficult.
- **Search LWE:** For truly uniform random samples from $\mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m$. Distinguishing between LWE's pair and uniform distribution is difficult.

3. Primitives

3.1. Oblivious transfer

The 1-out-of -2 OT protocol is a communication privacy protocol commonly used to protect the exchange of sensitive data, wherein a sender provides two secret values, and the receiver selects one without the sender being aware of the choice. The receiver chooses the desired secret value, while the sender remains oblivious to which one was selected. In the proposed protocol, the UC-secure OT protocol is employed.

3.2. A circuit f for Hamming distance

Given two strings $pw_0, pw_1 \in \{0,1\}^n$, the Hamming distance denotes the number of different bits, between n bits of the two strings. The definition is as follows:

$$a(pw_0, pw_1) = |\{j | pw_0[j] \neq pw_1[j], j \in \{n\}\} \tag{7}$$

f uses **XOR** to compute two strings and then generates a bit list to represent the difference between the two strings. Based on the bit list, it outputs 1 if the number of 0 is not less than $n - \tau$, and 0 otherwise (shown in Fig. 1).

In the protocol of this paper, f plays an important role. First, it does not encode the two input strings, but the result of the string comparison. For example, if $\mathcal{P}_o$ inputs a string pw_0 whose jth bit is 0, it puts this bit in the first bit; on the contrary, if this bit is 1, it inverts the label. Then, using the OT protocol, $\mathcal{P}_1$ retrieves the label corresponding to the jth bit of its string(pw_1). If the two jth bits are equal, he retrieves the 0 label, otherwise he retrieves the 1 label. As shown in Fig. 1, a threshold gate with n inputs is to be calculated. This gate returns 0 if the sum of the inputs is above some threshold (at least $n - \tau$ password string characters are different) and 1 otherwise.

3.3. Randomized fuzzy equality

The proposed fPAKE protocol introduces RFE protocol based on fPAKE protocol [1], where RFE protocol combines the garbled scheme and the UC-secure OT protocol.

$\mathbf{Gb}(1^{\lambda}, f) \rightarrow (F, s, a)$: The algorithm inputs security parameters and a circuit f and then outputs a garbled circuit F, an encrypted message s and a decrypted message a.

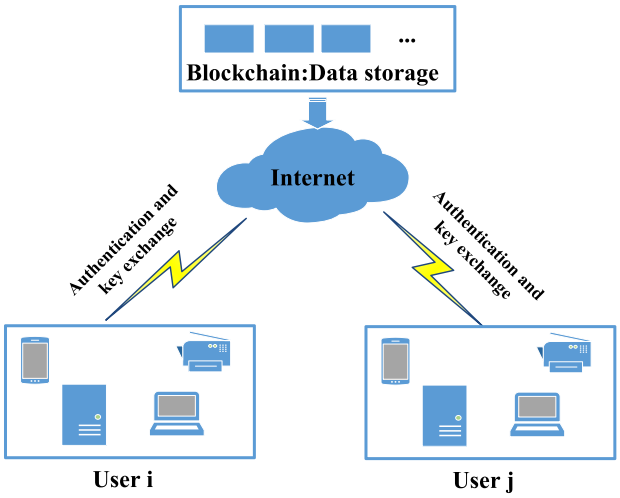


Fig. 2. System model of our protocol.

En$(e, x) \rightarrow X$: The algorithm inputs an encrypted message and a secret value x and outputs a garbled input X.

Ev$(F, X) \rightarrow Y$: The algorithm inputs a garbled circuit F and a garbled input X, and outputs a garbled output Y.

De$(a, Y) \rightarrow y$: The algorithm inputs a decrypted message and a garbled output Y and outputs plaintext y.

The garbling scheme has the following properties. *obliviousness*: if a garbled circuit F and a garbled input X a cannot reveal any information about a, then the scheme is oblivious; *privacy*: if it is known that decrypting a message a will give information about y but no more information about x, then the scheme is privacy; *authenticity*: if an adversary (given a garbled circuit F and a garbled input X) does not get $Y' \neq \mathrm{Ev}(F, X)$, then the scheme is authentic. The garbling scheme is projective if an encrypted message s consists of $2n$ labels, each of which is a random string. Each bit of the input is associated with two labels: one label indicating that the bit is a 0 event and one label indicating that the bit is a 1 event. And the RFE uses this feature to implement the Hamming distance calculation for both parties' passwords. More formally, suppose $\mathcal{P}_0$ outputs two labels $a_0[0]$ and $a_0[1]$; $a_1[0]$ and $a_1[1]$ are the two labels output by $\mathcal{P}_1$. Let Y_0 is the output label of $\mathcal{P}_1$ as the evaluation result, where $Y_0 \in (a_0[0], a_0[1])$. Let $Y_1 \in (a_1[0], a_1[1])$ is the output label of $\mathcal{P}_0$. Assume that $\mathcal{P}_0$ and $\mathcal{P}_1$ can safely compare $(a_0[1], Y_1)$ to $(Y_1, a_1[1],)$; if the comparison is successful, the output is1, otherwise 0 is output.

4. System model and universal composable security model

4.1. System model

As shown in Fig. 2, our system model consists of users and blockchain. We can see that the interaction between the two parties does not depend on a trusted third party. The user interacts with a device, which can be a smartphone, a computer, etc. After authentication between the two parties over the Internet, a shared key is obtained. The interaction between the two parties is securely stored on the blockchain.

4.2. UC framework

Universal Composable (UC) [30] has an important feature of the UC security model is composability. This composition property makes the design of security protocols more flexible and scalable. By analyzing the UC security model, it is possible to prove the security of the entire composite protocol without having to prove the security of the composite protocols individually. There are a number of UC-secure protocols [31–33] available that are used in networking scenarios. The basic assumption of the UC-secure model is the existence of a powerful attacker, i.e. an attacker with unlimited computing power and network access. Under the assumption, the model provides a guarantee of strong security. It protects against various attacks such as information leakage, manipulation and denial.

In the security analysis of the UC model, the main distinction is whether the protocol interacts with the ideal world or with the real world. In the real world, which is mainly the interaction between the adversary $\mathcal{A}$, the real protocol and the environment ($\mathcal{Z}$). Conversely, in the ideal world, which is mainly the interaction between the simulator Sim, the ideal protocol and the environment. If there is no Evn that can distinguish between real and ideal protocols with a non-negligible advantage, then real protocols have UC-secure.

4.3. Ideal functionality

In the security proof, we assume the existence of a simulator Sim that can emulate the interaction between two parties without knowledge of any password information. The simulator emulates the protocol in the real world and must remain indistinguishable from the real execution in the environment's view. Sim manipulates the adversary $\mathcal{A}$, interfering with the interaction between the two parties, particularly in the context of both active and passive attacks. The ideal functionality $\mathcal{F}_{\mathrm{fPAKE}}$ of fPAKE [25] models the interaction between the simulator and the parties. The indistinguishability between the real-world and ideal-world executions is demonstrated through a series of *Games*.

5. The proposed fPAKE protocol

5.1. fPAKE protocol

As shown in Fig. 3, the proposed fPAKE protocol is divided into two phases, the first phase is for the authentication of the fuzzy passwords between the parties and the second phase is for the key exchange between the parties. The proposed fPAKE protocol primarily employs **lossy** encryption schemes [25]. The CCA-secure key encapsulation mechanism, based on **lossy** encryption schemes, involves key generation, encryption, and decryption algorithms, all of which rely on the LWE problem. The CCA-secure public key encryption scheme (**DEnc**) utilized in the protocol meets the required security properties. In more detail, the proposed fPAKE protocol is briefly described below:

(1) $\mathcal{P}_i$ uses the OT protocol to generate the garbled input X for the passwords, where X_1 and X_0^* are generated using the encrypted message s_1 from $\mathcal{P}_1$; X_0 and X_1^* are generated using the encrypted message s_0 from $\mathcal{P}_0$. Thus, in the evaluation phase, there are two random strings as output labels, where one as "1" ($k_{i,correct} = a_i[1]$) and the other as "0" ($k_{i,wrong} = a_i[0]$). $\mathcal{P}_i$ evaluates the garbled input X of $\mathcal{P}_{1-i}$ and $\mathcal{P}_i$. Therefore, $\mathcal{P}_i$ obtains the garbled output $\mathrm{Y}_{1-\mathrm{i}}$ ($\mathrm{Y}_{1-\mathrm{i}} \in (k_{1-i,correct}, k_{1-i,wrong})$).

(2) The parties interact based on the **Loosy** and **CEnc** algorithms. $\mathcal{P}_0$ substitutes (X_0, X_1) in the **LoosyKeyGen** to generate a public/private key pair (pk, sk). $\mathcal{P}_1$ generates the ciphertexts (c, C) using **LoosyEnc** and **CEnc**. $\mathcal{P}_0$ evaluates ciphertexts and then generates the σ' for $\mathcal{P}_1$.

(3) $\mathcal{P}_i$ outputs SK.

Lossy.Setup(1^λ). The algorithm performs setup parameters. Using algorithm to generate public parameters and trapdoor.

- $(\mathbf{A}, T_A) \leftarrow TrapGen(1^\lambda)$.
- Output $(pp = \mathbf{A}, td = T_A)$.

LossyKeyGen. The algorithm performs key generation. Using algorithm to generate public key and private key.

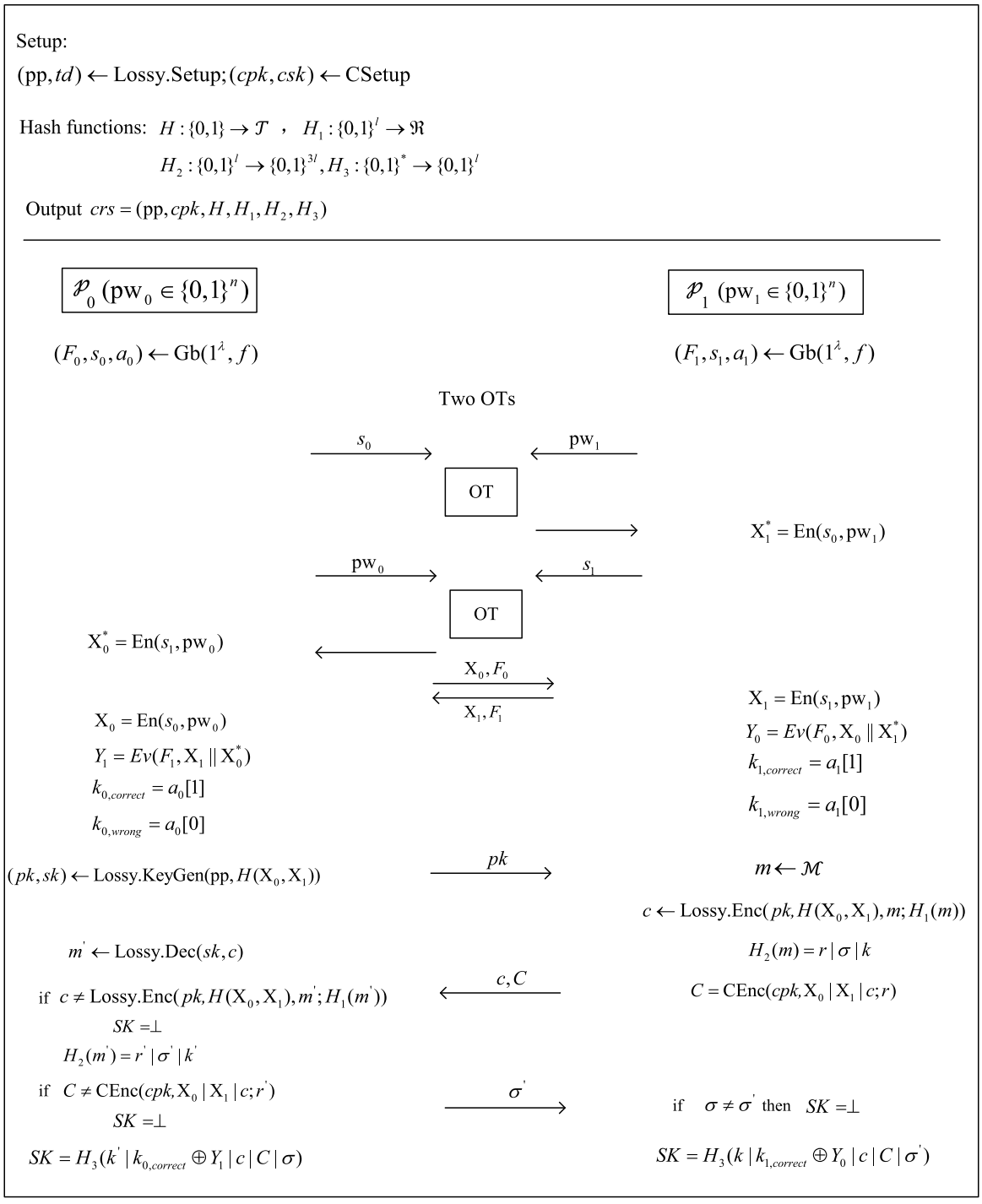


Fig. 3. Construction of the proposed fPAKE protocol.

- Input $\mathbf{s} \leftarrow \mathbb{Z}_q^n, \mathbf{e} \leftarrow D_{\mathbb{Z}^m,\partial}$, $\mathbf{v} \in \mathbb{Z}_q^m$; (Randomly selected $\mathbf{s}$ and $\mathbf{v}$, and $\mathbf{e}$ is selected from Gaussian distribution (6))
- output $(pk = A^T\mathbf{s} + \mathbf{e} - \mathbf{v}, sk = \mathbf{s})$.

LossyEnc. The algorithm performs encryption.

- Input $pk, \mathbf{v}, m \in \{0,1\}^\lambda$.
- Let $\mathbf{r} = pk + \mathbf{v}$, $\mathbf{e_i} \leftarrow D_{\mathbb{Z}^m,r}$, and $\mathbf{u}_i = \mathbf{A}\mathbf{e}_i$, $c_i = \mathbf{r}^{\mathrm{T}}\mathbf{e}_i + m_i q/2$. Output ciphertext $ct = (\mathbf{u}_1 | \cdots | \mathbf{u}_\lambda, c_1 | \cdots | c_\lambda)$, where $i \in \{1, \ldots, \lambda\}$. (The ciphertext ct is composed of $\mathbf{u}_i$ and $\mathbf{e}_i$.)

LossyDec. The algorithm performs decryption.

- Input $sk = \mathbf{s}, ct = (\mathbf{u}_1 | \cdots | \mathbf{u}_\lambda, c_1 | \cdots | c_\lambda)$.
- Let $d_i = c_i - \mathbf{s}^{\mathrm{T}}\mathbf{u}_i$, where $i \in \{1, \ldots, \lambda\}$.
- If $d_i \in [q/4, 3q/4]$, let $m_i = 1$, otherwise $m_i = 0$. Output $m = m_1 | \cdots | m_\lambda$.

IsLossy$(td = \mathbf{T_A}, pk, \mathbf{v})$. The algorithm performs verification.

- If **IsMessy**$(\mathbf{T_A}, \mathbf{A}, pk + \mathbf{v}) = messy$, output 1, otherwise output 0.

5.2. Security proof

$\mathbf{Real}_{\mathcal{Z},\mathcal{A}}$ denotes the interaction between the environment and the parties and the adversary in the real world. $\mathbf{Ideal}_{\mathcal{Z},Sim}$ denotes the

interaction between the environment and the simulator in the ideal world. We need to show that $\left|\Pr\left[\mathbf{Real}_{\mathcal{Z},\mathcal{A}} \Rightarrow 1\right] - \Pr\left[\mathbf{Ideal}_{\mathcal{Z},Sim} \Rightarrow 1\right]\right|$ is negligible after a series of Games. The Games are indistinguishable from each other in the environment. The security proof in this paper refer to [25] (Section 3.2). However, the inclusion of the RFE protocol in our construction necessitates a distinct security proof.

Game 0.

This is the real world $\mathbf{Real}_{\mathcal{Z},\mathcal{A}}$. Where the environment runs the protocol π (as shown in [25]) to see the interactions between the parties $\mathcal{P}_0$ $\mathcal{P}_1$, and the adversary, and the parties have access to the ideal functions of the OT. Here we have $Pr\left[\mathbf{Real}_{\mathcal{Z},\mathcal{A}} \Rightarrow 1\right] = Pr\left[Game_0 \Rightarrow 1\right]$

Game 1. (simulations of $\mathrm{X_i}$ and X_i^* for parties $\mathcal{P}_i$ in case of passive attacks)

In $Game1$, the simulator emulates $\mathrm{X_i}$, X_i^* and then runs according to the protocol π to implement the interaction with $\mathcal{P}_0$ and $\mathcal{P}_1$, where $\mathrm{X_i} \in \{\mathrm{X_1}, \mathrm{X_0}\}$, $\mathrm{X_i^*} \in \{\mathrm{X_1}^*, \mathrm{X_0}^*\}$, $\mathcal{P}_i \in \{\mathcal{P}_1, \mathcal{P}_0\}$, $a(pw_0, pw_1) > \partial$. In this Game, $\mathrm{X_i}$ is generated differently than in $Game0$. The $\mathrm{X_i}$ is by garbled output randomness simulator on $(f, (pw_0, pw_1))$ to get random sets $(F_0, \mathrm{X_0}, X_0^*)$ and $(F_1, \mathrm{X_1}, X_1^*)$. Chosen $b \in \{0,1\}$ randomly, if $b = 0$, then output real instances $(F_i, \mathrm{X_i}, X_i^*)$, and the view of the environment in $Game0$; if $b = 1$, then a simulator generated $(F_i, \mathrm{X_i}, X_i^*)$ is output, and the view of the environment in $Game1$. So we have $\left|\Pr\left[Game_1 \Rightarrow 1\right] - \Pr\left[Game_0 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 2. (simulation for crs)

The difference is that in $Game1$ the hash function is simulated by the simulator as random oracles in the parameter setting phase of the protocol. For hash function H, H_1, H_2, H_3, Sim implements the hash function by simulating random oracles. For parameter generation, where Sim is honestly generated, apply $(pp, td) \leftarrow Lossy.Setup; (cpk, csk) \leftarrow CSetup$. So we have $\left|\Pr\left[Game_2 \Rightarrow 1\right] - \Pr\left[Game_1 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 3. (simulation of $r|\sigma|k$ for party $\mathcal{P}_1$ instances and simulation of σ' for party $\mathcal{P}_0$ instances in case of passive attacks)

It is roughly the same as $GameG3$.

- For $\mathcal{P}_1$, after receiving the pk, instead of using $H_2(m) = r|\sigma|k$, $r|\sigma|k \leftarrow \{0,1\}^{3\lambda}$ is randomly selected.
- For the (c, C) received by $\mathcal{P}_0$, we know that this is the $r|\sigma|k$ generated by Sim and used to generate (c, C). So here it is straightforward to generate SK from Sim and output $\sigma' = \sigma$.

According to [31], we know that $LossyEnc$ is CCA-secure, so in fact for a PPT adversary access to decryption oracle, an indistinguishable (c, C) can be generated. So we have $\left|\Pr\left[Game_3 \Rightarrow 1\right] - \Pr\left[Game_2 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 4. (simulation of SK in case of passive attacks)

- For $\mathcal{P}_0$, unlike in $Game3$, here Sim randomly chooses $SK \leftarrow \{0,1\}^{\lambda}$ and then stores SK, where (c, C) must be generated from Sim and $r|\sigma|k$ is also randomly generated from Sim. And for σ' Sim returns the same result as in $Game3$.
- For $\mathcal{P}_1$, Sim compares σ' to σ, where σ is generated from $\mathcal{P}_1$. If $\sigma' = \sigma$, Sim updates the SK store for $\mathcal{P}_0$ while sending the same SK to $\mathcal{P}_1$.

According to [25], in the view of the adversary $\mathcal{A}$. $r|\sigma|k$ is chosen at randomly. In $Game3$, SK is statistically uniform distribution, so for $\mathcal{Z}$, $Game4$ is the same as $Game3$. So we have $\left|\Pr\left[Game_4 \Rightarrow 1\right] - \Pr\left[Game_3 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 5. (simulations of $\mathrm{X_i}$ for parties $\mathcal{P}_i$ in case of active attacks)

In $Game5$, due to an active attack, the $d(\mathrm{pw_0}, \mathrm{pw_1}) \leq \partial$. In $Game5$, the input $\mathcal{P}_i$ of pw'_{1-i} to the ideal OT functionality, and then query on garbled output randomness simulator to output $\mathcal{P}_i$. Sim will output r as $k_{i,correct}$. Randomly pick a $b \in \{0,1\}$, if $b = 1$, the real $k_{i,correct}$ is output as r, and the view of the environment is in $Game4$; if $b = 0$, a random value is output as r, and the view of the environment is in $Game5$. So we have $\left|\Pr\left[Game_5 \Rightarrow 1\right] - \Pr\left[Game_4 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 6. (simulation for $\mathcal{P}_0$ instances in case of active attacks)

In $Game6$, add a rejection rule here, otherwise same as $Game5$.

- For $\mathcal{P}_0$, when $(\tilde{c}, \tilde{C})$ is received, since it is an active attack, we know that $(\tilde{c}, \tilde{C})$ is not an instance generated by $\mathcal{P}_1$. Let pw_i be the password of $\mathcal{P}_i$ and $\mathrm{X_i}$ be generated from Sim.
- **Rejection rule:** For received $(\tilde{c}, \tilde{C})$, Sim firstly generates $\mathrm{X_0}'|\mathrm{X_1}'|c' \leftarrow CDec(csk, \tilde{C})$. If $\mathrm{X_0}'|\mathrm{X_1}' \neq \mathrm{X_0}|\mathrm{X_1}$ or$c' \neq \tilde{c}$, then Sim rejects $(\tilde{c}, \tilde{C})$ and outputs $SK = \perp$.

The difference between $Game5$ and $Game6$ is this rejection rule. In $Game5$ there is no such requirement. We have $\tilde{C} = CEnc(cpk, \mathrm{X_0}|\mathrm{X_1}|\tilde{c}; r)$, $\mathrm{X_0}'|\mathrm{X_1}'|c' \leftarrow CDec(csk, \tilde{C})$, but $\mathrm{X_0}'|\mathrm{X_1}'c' \neq \mathrm{X_0}|\mathrm{X_1}|\tilde{c}$. So in $Game5$ $(\tilde{c}, \tilde{C})$ will also result in a rejection. So we have $\left|\Pr\left[Game_6 \Rightarrow 1\right] - \Pr\left[Game_5 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 7. (get rid of SK in simulation for $\mathcal{P}_0$ in case of active attacks)

In order to respond to an $\mathcal{A}$'s query, the simulator Sim, without knowing the SK and needs to generate a m'. The difference with $Game6$ is that the Sim generates m' differently. For $\mathcal{P}_0$, it received $(\tilde{c}, \tilde{C})$ must be forged by the adversary, so Sim does not get $m' \leftarrow \mathrm{Lossy.Dec}(sk, \tilde{c})$. Therefore Sim has to make a random oracle query.

- If (m', r_H) in random oracle such that $\tilde{c} \leftarrow \mathrm{Lossy.Enc}(pk, H(\mathrm{X_0}, \mathrm{X_1}), m'; r_H)$, then the result is the same as $Game6$, both $m' \leftarrow \mathrm{Lossy.Dec}(sk, \tilde{c})$.
- If (m', r_H) **not** in random oracle such that $\tilde{c} \leftarrow \mathrm{Lossy.Enc}(pk, H(\mathrm{X_0}, \mathrm{X_1}), m'; r_H)$, then show that $H_1(m')$ is random. In $Game6$ and $Game7$, Sim outputs $SK = \perp$.

We have $\left|\Pr\left[Game_7 \Rightarrow 1\right] - \Pr\left[Game_6 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 8. (simulation for $\mathcal{P}_0$'s messages without pw_0) In $Game8$ the Sim is the same as in $Game7$ except that it simulates $F_0, \mathrm{X_0}$ for $\mathcal{P}_0$. And for $F_1, \mathrm{X_1}$ for $\mathcal{P}_0$ follow the normal steps to generate. If both parties are honest, the simulator does not need to simulate anything related to OT, and the environment cannot observe any functional inputs or outputs of OT. The simulator uses the obliviousness simulator to generate $F_0, \mathrm{X_0}$ whose distribution is random and indistinguishable from $Game7$. So we have $\left|\Pr\left[Game_8 \Rightarrow 1\right] - \Pr\left[Game_7 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 9. (simulation for $\mathcal{P}_1$'s messages without pw_1) In $Game9$ the Sim is the same as in $Game8$ except that it simulates $F_1, \mathrm{X_1}$ for $\mathcal{P}_1$. The simulator uses the obliviousness simulator to generate $F_1, \mathrm{X_1}$ whose distribution is random and indistinguishable from $Game8$. So we have $\left|\Pr\left[Game_9 \Rightarrow 1\right] - \Pr\left[Game_8 \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 10. (simulation of generating c)

In $Game10$ the Sim is the same as in $Game9$ except that it simulates (c, C) for $\mathcal{P}_1$.

- **passive attacks on $\mathcal{P}_1$.** For receiving pk, Sim will randomly select c from the $c \leftarrow CT$ instead of computing $c \leftarrow \mathrm{Lossy.Enc}(pk, H(\mathrm{X_0}, \mathrm{X_1}), m; H_1(m))$. Here the randomly selected c and $Game9$ are randomly indistinguishable. Note that C are computed here in the same way as in $Game9$.
- **active attacks on $\mathcal{P}_1$.** For receiving $\tilde{pk}$, it must be forged by the adversary, so consider three cases here

  1. If $((\mathrm{X_0}', \mathrm{X_1}'), r_H)$ **not** in random oracle such that $0 \leftarrow \mathrm{IsLossy}(td, \tilde{pk}, r_H)$. Thus for the adversary, $H(\mathrm{X_0}, \mathrm{X_1})$ is random. So we can pick a random c, the same as $Game9$.
  2. $H(\mathrm{X_0}', \mathrm{X_1}')$ is a label for pk. If $(\mathrm{X_0}', \mathrm{X_1}')$ is not the correct information, then $H(\mathrm{X_0}, \mathrm{X_1})$ is a lossy label for pk. If $(\mathrm{X_0}, \mathrm{X_1})$ is in a random oracle, then $1 \leftarrow \mathrm{IsLossy}(td, \tilde{pk}, r_H\prime)$, so $H(\mathrm{X_0}, \mathrm{X_1})$ is a lossy label for $\tilde{pk}$. Otherwise, $H(\mathrm{X_0}, \mathrm{X_1})$ is random for the adversary. So we can choose a random c.
  3. If$((\mathrm{X_0}, \mathrm{X_1}), r_H)$, $((\mathrm{X_0}', \mathrm{X_1}'), r_H)$ in random oracle such that $\mathrm{IsLossy}(td, \tilde{pk}, r_H) = \mathrm{IsLossy}(td, \tilde{pk}, r_H\prime) = 0$. Here Sim directly aborts the process.

Therefore, we have $\left|\Pr\left[Game_{10} \Rightarrow 1\right] - \Pr\left[Game_{9} \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 11. (**simulation of generating** C)

In $Game11$ the Sim is the same as in $Game10$ except that Sim simulation generates c, C for $\mathcal{P}_0$. The analysis here is the same as $Game10$, with a randomly chosen r to generate C, otherwise the same as $Game10$. So we have $\left|\Pr\left[Game_{11} \Rightarrow 1\right] - \Pr\left[Game_{10} \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 12. (**simulation of** $\tilde{\sigma}$ **for** $\mathcal{P}_1$ **in case of active attacks**)

In $Game12$ the Sim is the same as in $Game11$ except that it simulates the generation of SK after receiving $\tilde{\sigma}$. In the same process as $Game10$ above, Sim directly outputs $SK = \perp$ when $(X_0', X_1') \neq (X_0, X_1)$. Here we consider the output when $\tilde{\sigma} = \sigma$. But we know that the choice of r is random and independent of the other elements, so the probability of the adversary guessing correctly is negligible. So we have $\left|\Pr\left[Game_{12} \Rightarrow 1\right] - \Pr\left[Game_{11} \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Game 13. (**Integration of** Sim **with** $\mathcal{F}_{\text{fPAKE}}$)

In $Game13$ the Sim is the same as in $Game12$ except that Sim accesses $\mathcal{F}_{\text{fPAKE}}$. Sim determines whether (X_0', X_1') is equal to (X_0, X_1) by way of $\mathcal{F}_{\text{fPAKE}}$. So we have $\left|\Pr\left[Game_{13} \Rightarrow 1\right] - \Pr\left[Game_{12} \Rightarrow 1\right] \leq negl(\lambda)\right|$.

Sim can be done without passwords, so $Game13$ is $\mathbf{Ideal}_{\mathcal{Z},Sim}$.

Finally, through the above series of $Games$, we conclude that:

$$\left|\Pr\left[\mathbf{Real}_{\mathcal{Z},\mathcal{A}}\right] - \Pr\left[\mathbf{Ideal}_{\mathcal{Z},Sim}\right]\right| \leq negl(\lambda).$$

6. Performance analysis

In this subsection, this work compares the proposed fPAKE protocol with other protocols in terms of functionality and cost, and specific analyses of computation and storage costs are provided.

6.1. Comparison of functionality

As shown in Table 3, we compare the proposed fPAKE protocol with other protocols, the table examines whether the security of each protocol achieves UC-secure and whether the security proofs are conducted under the Random Oracle (RO) model or the Standard model. Additionally, the security of the algorithms achieved in the protocols is compared, specifically whether the algorithm's security provides chosen plaintext attack (CPA) security or chosen ciphertext attack (CCA) security. The $*$ indicates the aspect is not discussed in the paper. Obviously, only proposed fPAKE protocol resists quantum attacks, among other properties. Lyu et al.'s protocol [25] and Song et al.'s protocol [34] are lattice-based schemes, the protocols in [1,35,36], and [37] are based on bilinear pairings. The protocols in [1,36], and the proposed fPAKE protocol support fuzzy password authentication. However, the protocol in [1] lacks resistance to quantum attacks, and the protocol in [36] provides neither UC security nor quantum security. The protocols in [25,35,37], and [34] do not achieve fuzzy password authentication. Where the security proof of protocols in [35,37] are under the RO model, but neither achieves UC-secure nor quantum security. The protocols in [25,34] exhibit quantum security, the security proof of protocol in [34] is under the Standard model, and ensures CPA security for interacting parties. The protocol in [25] achieved UC-secure under the RO model. In summary, the proposed protocol outperforms other protocols in terms of security and performance.

6.2. Comparison of cost

In Table 4, only schemes based on the LWE hard problem are compared. Communication and storage costs are analyzed with schemes [25,34]. It is assumed that the identity size is 160 bits, the outputs of hash functions H and H_1 are 160 bits, and the label is 60 bits. Additionally, the outputs of hash functions H_2 and H_3 are 512 and 256 bits, respectively. For the LWE instance $(\boldsymbol{A}, \boldsymbol{c}) = (\boldsymbol{A}, \boldsymbol{A}^{\mathrm{T}}\boldsymbol{s}+\boldsymbol{x}) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m$, let $\boldsymbol{A}$ is 512 bits, $\boldsymbol{s}$ and $\boldsymbol{x}$ is 320 bits,respectively.

Table 3
Comparison of functionality.

Fuzzy	Protocol	UC-secure	Model	Assumption	Security
×	[35]	×	RO	DDH	*
×	[37]	×	RO	CDH	*
×	[25]	✓	RO	LWE	CCA
×	[34]	×	Standard	RLWE	CPA
✓	[1]	✓	*	CDH	*
✓	[36]	×	RO	CDH	CCA
✓	Our protocol	✓	RO	LWE	CCA

Table 4
Comparison of communication and storage cost.

Fuzzy	Protocol	Communication cost (bit)	Storage cost (bit)
×	Lyu et al's protocol [25]	1216	1980
×	Song et al's protocol [34]	2464	2080
✓	our protocol	1456	1920

In protocol [25], the storage of information is represented as $pwfile$, and the exchange of information includes (pk, C, c, σ'). Thus, the communication cost and storage cost are $320 + 320 + 320 + 256 = 1216$ bits and $512 + 320 + 160 + 160 + 512 + 256 + 60 = 1980$ bits, respectively. In protocol [34], the storage involves (pk_0, pk_1, τ), and the exchange of information includes (M_0, M_1), where $M_0 = (Aid, b_0, c_0, n_0, hc_c, s)$, and $M_1 = (b_1, w_1, c_1, n, hs_s)$. The communication cost and storage cost for protocol [34] are $160+160+160+320+512+160+160+320+512 = 2464$ bits and $(320+320+320)+(320+320+320)+160 = 2080$ bits, respectively. In the proposed protocol, the storage involves crs and the exchange of information is $(X_0, X_1, F_0, F_1, pk, C, c, \sigma')$. The communication and storage costs for the proposed protocol are $(60 + 60) + (60 + 60) + 320 + 320 + 320 + 256 = 1456$ bits and $512 + 320 + 160 + 160 + 512 + 256 = 1920$ bits, respectively. As shown in Table 4, the proposed protocol achieves fuzzy password authentication while maintaining communication and storage efficiency.

Table 5 presents the computation cost incurred by the sender and receiver during the authenticated key phase, comparing the proposed fPAKE protocol with protocols [25,34]. $Hash$ denotes the time for a hash operation, $Sample$ refers to the time for a sampling algorithm over a lattice, and Exp represents the time for an arithmetic operation. Compared to protocol [34], the proposed protocol is efficient while achieving fuzzy password authentication. Compared to protocol [25], some efficiency is sacrificed to achieve fuzzy password authentication, allowing the protocol to be applied to a wider range of scenarios.

6.3. Experimental evaluation

The system configurations used for the implementation of the proposed protocol are as follows: The experiments are running on AMD Ryzen 7 7840HS with Radeon 780M Graphics 3.80 GHz CPU and 32 GB RAM. The compilation environment is C++ 14 and Microsoft Visual Studio Code 2022. Based on the NTL library, the computation cost under the security parameters $n = 64, 128, 256, 512$.

Fig. 4 illustrates a comparison of the computation cost among the proposed fPAKE protocol and Lyu et al.'s protocol [25], Song et al.'s protocol [34]. Fig. 4(a) depicts the computation cost of the sender, while Fig. 4(b) shows the computation cost of the receiver. As shown in Fig. 4(a) and (b), the computation cost of the proposed fPAKE protocol is slightly higher than that of Lyu et al.'s protocol and marginally lower than that of Song et al.'s protocol. To achieve fuzzy password authentication and UC security in the proposed fPAKE protocol, two rounds of the OT protocol are incorporated during the initial phase. As a result, the proposed protocol incurs slightly higher computational overhead compared to Lyu et al.'s protocol. Although the proposed fPAKE protocol does not have a computational advantage over Lyu et al.'s protocol [25], some efficiency is sacrificed to enable fuzzy password authentication. In the proposed fPAKE protocol, the sender

Table 5
Comparison of computation cost.

Fuzzy	Protocol	Authenticated key exchange		Total
		Sender($\mathcal{P}_0$)	Receiver($\mathcal{P}_1$)	
×	Lyu et al's protocol [25]	$2Hash + Sample + 3Exp$	$2Hash + 2Exp$	$4Hash + Sample + 5Exp$
×	Song et al's protocol [34]	$3Hash + 5Sample + 3Exp$	$5Hash + 3Sample + 5Exp$	$8Hash + 8Sample + 8Exp$
✓	our protocol	$2Hash + 2Sample + 6Exp$	$2Hash + Sample + 5Exp$	$4Hash + 3Sample + 11Exp$

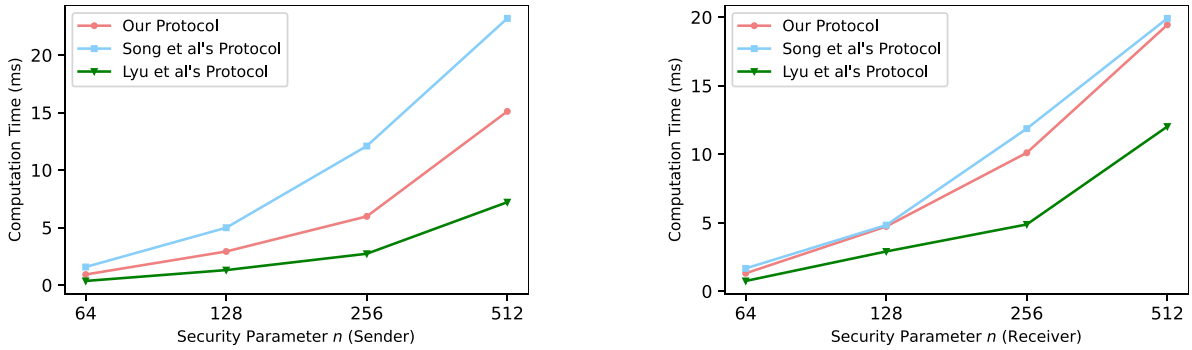


Fig. 4. Comparison of computation cost with different security parameters.

utilizes the **LossyKeyGen** and **LossyDec** algorithms, while the receiver utilizes the **LossyEnc** algorithm. Compared to Song et al.'s protocol, the proposed fPAKE protocol does not repeatedly invoke the sampling algorithm for interactions, thereby reducing computation time. Overall, the proposed fPAKE protocol efficiently implements fuzzy password authentication under the UC model, with computational overhead that remain competitive compared to other protocols.

In the simulations, it is assumed that the total number of participating Key Generation Centres (KGCs) is n, i.e. n = 20, and keys are securely generated for each user by the KGCs. A leader, referred to as a proposer or miner, is selected from the KGCs in a round-robin manner and is responsible for creating, validating, and adding blocks to the blockchain. Various fields in the blocks of the implemented blockchain are summarized as follows. The block version is 32 bits, the hash values H and H_1 from the previous block are 160 bits, and the label is 60 bits. The hash values H_2 and H_3 are 512 and 256 bits, respectively. The transaction details within the protocol are $(X_0, X_1, F_0, F_1, pk, C, c, \sigma')$, requiring $60+60+320+320+320+256 = 1456$ bits in total. Furthermore, the payload of the block is $1456t$ bits, where t is the number of transactions. The total block size is estimated to be $(1180 + 1456t)$ bits. The transactions per second (TPS) are calculated based on the time required to process different numbers of transactions per block. As shown in Fig. 5, with 30 fixed blocks, varying transaction values of 30, 40, 50, and 60 result in computation times of 237, 259, 288, and 310 s, respectively. The figure demonstrates that as the number of transactions per block increases, the TPS value proportionally increases.

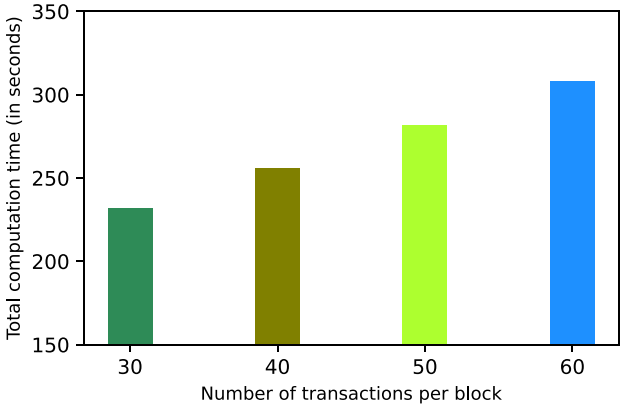


Fig. 5. Computation time for varied Number of transactions per block.

7. Application

In the proposed fPAKE protocol, parties act as both data owners and data users, making it highly suitable for applications in the IoV. The evolution of IoV has enabled vehicles to interact and offer services, transforming them from simple vehicles into intelligent terminals. However, the openness also exposes vehicles to potential malicious attacks and privacy breaches. To address this, a key authentication protocol combined with a blockchain system is considered for IoV applications. Blockchain serves as an interaction platform, eliminating the need for centralized authorities and reducing the risk of information leakage.

As shown in Fig. 6, when a vehicle interacts with a data center in an internet environment, the operational data between both parties is stored on the blockchain, ensuring the privacy of the vehicle. First, an interaction request is initiated by the vehicle $\mathcal{P}_1$, with its information stored on the blockchain and sent to the data center $\mathcal{P}_0$. If the passwords between the two parties meet the protocol requirements, the interaction proceeds. The data center password is pw, and the set of vehicle passwords is $U = (pw_1, pw_2, \ldots pw_n)$. After the interaction request is established, both parties communicate according to the protocol. Prior to the interaction, system parameters information $(pp, cpk, H, H_1, H_2, H_3)$ is generated and stored on the blockchain. And then the message $(X_0, X_1, F_0, F_1, pk, C, c, \sigma')$ is exchanged between the parties. Upon completing the interaction, a shared key is generated between the two parties.

8. Conclusions

Considering that UC-secure protocols can be applied to complex blockchain scenarios, this work primarily combines the RFE protocol and the OT protocol to achieve the UC-secure fPAKE protocol. Based on the PAKE protocol, the subsequent encryption phase of the fPAKE protocol is designed, as well as the interaction between the parties. The encryption and decryption algorithms are resistant to quantum attacks, making the application process more security. Combined with a series of games under the UC model, this work provides a formal security

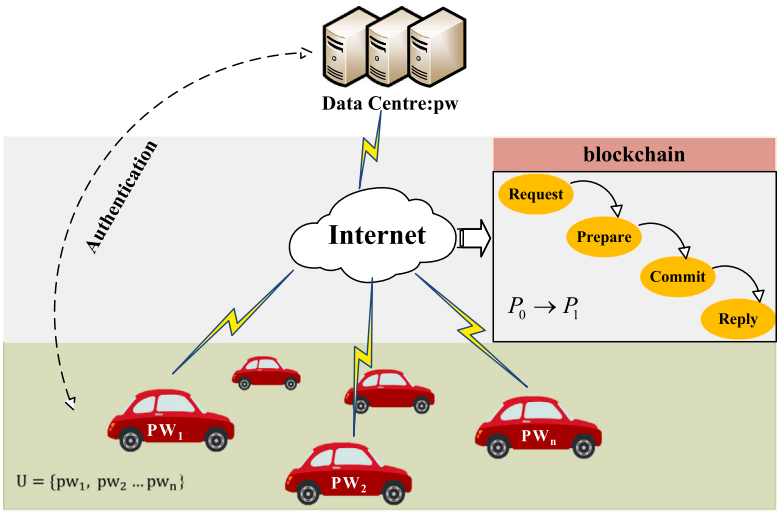


Fig. 6. Application scenarios of blockchain in IoV.

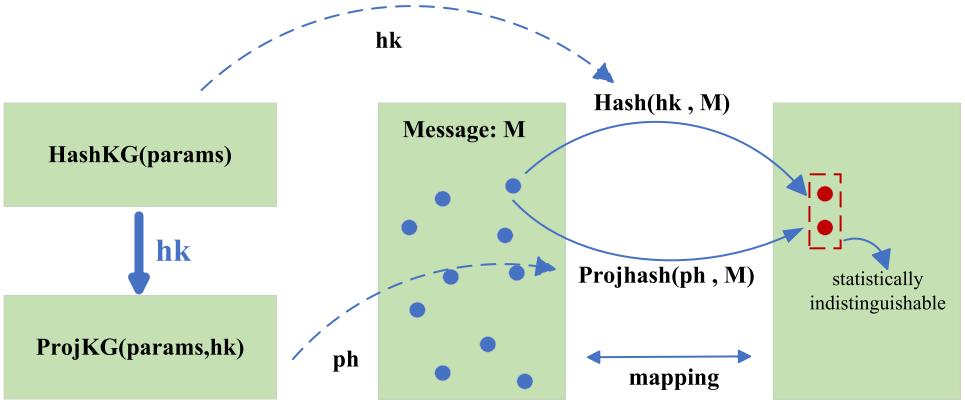


Fig. 7. Structure of SPHF.

proof. The proposed fPAKE protocol implements two-way authentication, which results in a greater number of interactions. Increased interactions consume more network resources and incur additional computation cost. However, the number of interactions between parties directly impacts the efficiency of the protocol. A lower number of interactions increases efficiency. Future work will focus on the SPHF [38]. One current challenge is how to implement interactions between parties using SPHF while ensuring fuzzy password authentication, specifically how to hash the fuzzy password. The following outlines the specific structure of SPHF:

- A hash key hk is first generated by $HashKG(params)$ and then a mapping result is generated by $Hash(hk, M)$. Where params are system parameters and M is message information.
- The hk that was generated in the previous part is needed in $ProjKG(params, hk)$. The function $ProjKG(pa$ $rams, hk)$ is used to generate hash key ph. A result of the mapping is then generated by $Projhash(ph, M)$.

As shown in Fig. 7, SPHF consists of two components. The first component includes $HashKG(params)$ and $Hash(hk, M)$, while the second component consists of $ProjKG(params, hk)$ and $Projhash(ph, M)$. Additionally, the mapping results generated by $Hash(hk, M)$ and $Projhash$ (ph, M) are statistically indistinguishable. This ensures security during interactions between parties. According to the definition of SPHF, it is evident that it consists of two interrelated hash functions with indistinguishable outputs. Thus, by introducing SPHF into the protocol, the number of interactions between parties can be reduced.

CRediT authorship contribution statement

Qihong Chen: Writing – original draft. **Changgen Peng:** Methodology. **Dequan Xu:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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